



US0D1092407S

(12) **United States Design Patent**
Sasaki et al.

(10) **Patent No.: US D1,092,407 S**
(45) **Date of Patent: ** Sep. 9, 2025**

(54) **OPTICAL FIBER CONNECTOR**

(71) Applicants: **SUMITOMO ELECTRIC INDUSTRIES, LTD.**, Osaka (JP);
SUMITOMO ELECTRIC OPTIFRONTIER CO., LTD.,
Yokohama (JP)

(72) Inventors: **Dai Sasaki**, Osaka (JP); **Masahiro Shibata**, Yokohama (JP)

(73) Assignees: **SUMITOMO ELECTRIC INDUSTRIES, LTD.**, Osaka (JP);
SUMITOMO ELECTRIC OPTIFRONTIER CO., LTD.,
Yokohama (JP)

(**) Term: **15 Years**

(21) Appl. No.: **29/938,377**

(22) Filed: **Apr. 19, 2024**

Related U.S. Application Data

(62) Division of application No. 29/849,367, filed on Aug. 10, 2022, now Pat. No. Des. 1,036,389.

(30) **Foreign Application Priority Data**

Feb. 24, 2022 (JP) 2022-003589 D
Feb. 24, 2022 (JP) 2022-003591 D

(51) **LOC (15) Cl.** **13-03**

(52) **U.S. Cl.**
USPC **D13/133**

(58) **Field of Classification Search**
USPC D6/682.6; D8/98, 349; D10/70, 109.1,
D10/109.2, 46, 66, 81; D13/133, 147,
(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,016,968 A 5/1991 Hammond et al.
5,838,856 A 11/1998 Lee
(Continued)

FOREIGN PATENT DOCUMENTS

JP 2015-025868 A 2/2015
JP 2017-156617 A 9/2017

OTHER PUBLICATIONS

FS MTP connectors, posted Sep. 27, 2024 [online], [retrieved May 2, 2025]. Retrieved from internet, <https://www.fs.com/blog/a-comprehensive-guide-to-mtp-connector-958.html> (Year: 2024).*

(Continued)

Primary Examiner — George D. Kirschbaum

Assistant Examiner — Denis Houyoux

(74) *Attorney, Agent, or Firm* — Oliff PLC

(57) **CLAIM**

The ornamental design for an optical fiber connector as shown and described.

DESCRIPTION

FIG. 1 is a front view of an optical fiber connector of our new design;

FIG. 2 is a rear view of the optical fiber connector of FIG. 1;

FIG. 3 is a top plan view of the optical fiber connector of FIG. 1;

FIG. 4 is a bottom plan view of the optical fiber connector of FIG. 1;

FIG. 5 is a right side view of the optical fiber connector of FIG. 1;

FIG. 6 is a left side view of the optical fiber connector of FIG. 1;

FIG. 7 is a front, top plan and left side perspective view of the optical fiber connector of FIG. 1;

FIG. 8 is a front, bottom plan and left side perspective view of the optical fiber connector of FIG. 1;

FIG. 9 is a rear, top plan and right side perspective view of the optical fiber connector of FIG. 1;

FIG. 10 is a sectional view taken along the line 10-10 of the optical fiber connector of FIG. 6;

FIG. 11 is an enlarged view defined by the lines 11-11 and 11'-11' of the optical fiber connector of FIG. 8;

(Continued)

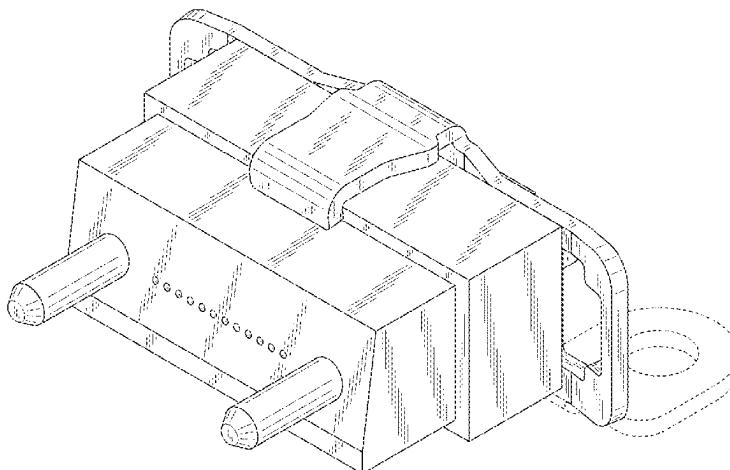


FIG. 12 is an enlarged view defined by the lines 12-12 and 12'-12' of the optical fiber connector of FIG. 8; FIG. 13 is an enlarged view defined by the lines 13-13 and 13'-13' of the optical fiber connector of FIG. 9; and, FIG. 14 is an environmental perspective view showing the optical fiber connector of FIG. 1.

The features shown in evenly-dashed broken lines depict environmental subject matter only and form no part of the claimed design. The dot-dash broken lines in the drawings represent the bounds of the claimed subject matter, the dot-dash broken lines, themselves forming no part thereof.

1 Claim, 14 Drawing Sheets

(58) Field of Classification Search

USPC D13/154, 156, 146, 101, 118, 123;
D14/433, 420, 256, 428, 206, 372, 426,
D14/429, 435, 453, 240, 242, 357, 358,
D14/140-140, 155, 125, 137, 139, 243,
D14/348, 349, 351, 354, 355, 251, 253,
D14/15, 10, 122, 144; D16/134, 130,
D16/131, 132, 135, 136, 202-208, 210,
D16/245, 219, 221, 235, 237, 313, 314,
D16/326; D22/109, 108; D23/40, 43,
D23/260; D24/120, 113, 118, 119, 121,
D24/122, 124, 125, 128, 137, 186, 172,
D24/129; D26/24, 28, 122, 113, 118;
D32/35
CPC A45C 11/182; A45C 13/002; A45F 5/00;
H04M 1/21; H04M 1/04; F16B 47/003;
F16M 13/04; H01R 13/443; H01R
33/965; H01R 13/52; H01R 2201/16;
Y10T 29/53235; G02B 6/38875; G02B
6/3888; G02B 6/3825; G02B 6/3893;
G02B 6/406; G02B 6/3895; G02B 6/387;
G02B 27/026; G02B 23/16; G02B
2003/0093; G02B 3/04; G02B 6/3831;
G02B 6/3887; G02B 6/3866; G02B
6/3871; G02B 6/3898; G02B 6/3869;
G02B 6/3882; G02B 6/3885; G02B
6/3846; G02B 6/3802; G02B 6/3865;
G02B 6/381; G02B 6/3644; G02B 6/32;
G01C 3/00; G01C 3/02; G01C 3/04;
G01C 3/06; G01C 3/08; G01C 3/085;
G01C 3/10; G01C 3/12; G01C 3/16;
G01C 3/18; G01C 3/20; G01C 3/22;
G01C 3/24; G01C 3/26; G01C 3/28;
G01C 3/30; G01C 3/32; B60Q 1/04;
B60Q 1/26; F21S 8/026; F21S 8/04;
F21V 21/02; F21V 21/04; H04J 11/00;
H04J 13/00; E06B 9/24; G03B 17/02;
G03B 11/00; G03B 15/14; C03B
2215/46; H04L 41/0896; H04L 41/5022;

H04L 67/06; H04L 67/1097; H04L 67/34;
A61B 2090/306; A61B 90/35

See application file for complete search history.

(56)

References Cited

U.S. PATENT DOCUMENTS

6,530,696 B1	3/2003	Ueda et al.	
6,616,343 B2 *	9/2003	Katsura	C08K 7/08 385/60
D750,567 S *	3/2016	Katagiyama	D13/147
9,465,170 B1	10/2016	Childers et al.	
9,563,027 B2	2/2017	Childers et al.	
9,817,194 B2	11/2017	Childers et al.	
D828,811 S	9/2018	Aoshima et al.	
D828,812 S	9/2018	Aoshima et al.	
11,280,966 B2	3/2022	Cloud et al.	
11,719,892 B2	8/2023	Higley et al.	
D1,036,389 S *	7/2024	Sasaki	D13/133
2001/0043776 A1	11/2001	Shimoji et al.	
2012/0328244 A1 *	12/2012	Sasaki	G02B 6/403 385/59
2013/0183004 A1	7/2013	Hughes et al.	
2013/0266271 A1	10/2013	Li et al.	
2015/0185423 A1	7/2015	Matsui et al.	
2017/0176694 A1	6/2017	Childers et al.	
2019/0072726 A1	3/2019	Ohmura	
2019/0361177 A1	11/2019	Aoshima et al.	
2021/0215887 A1	7/2021	Cloud et al.	
2021/0325614 A1	10/2021	Childers et al.	
2022/0026645 A1 *	1/2022	Arao	G02B 6/3644
2023/0036226 A1 *	2/2023	Wang	G02B 6/3882
2023/0273377 A1 *	8/2023	Nakane	G02B 6/3885 385/57

OTHER PUBLICATIONS

Fujikura TMT ferrule, posted Mar. 3, 2024 [online], [retrieved May 2, 2025]. Retrieved from internet, <https://www.optic-product.fujikura.com/optical-components/en/products/tmt-ferrule-technology/> (Year: 2024).*

Glenair MT elite ferrule kits, posted May 24, 2024 [online], [retrieved May 2, 2025]. Retrieved from internet, <https://cdn.glenair.com/fiberoptics/mt-ferrule/pdf/181-133.pdf> (Year: 2024).*

Optico MT-FA fiber jumper, posted Nov. 17, 2022 [online], [retrieved May 2, 2025]. Retrieved from internet, <https://www.fiberopticom.com/patch-cord/mt-patch-cord/mt-fa-fiber-jumper.html> (Year: 2022).*

Sumitomo nanoplug mt, posted Aug. 10, 2022 [online], [retrieved May 2, 2025]. Retrieved from internet, <https://sumitomoelectric.com/products/NanoPlug> (Year: 2022).*

Feb. 15, 2024 Office Action issued in U.S. Appl. No. 29/849,375. 367 Sumitomo1 NPL, posted Oct. 16, 2021 [online], [retrieved Feb. 1, 2024]. Retrieved from internet, https://sumitomoelectric.com/Advanced_Multi-fiber_Connectors/AirMT (Year: 2021).

367 abd 375 Nanoplug NPL, posted Apr. 15, 2021 [online], [retrieved Feb. 1, 2024]. Retrieved from internet, <https://sumitomoelectric.com/products/NanoPlug> (Year: 2021).

May 13, 2025 Notice of Allowance issued in U.S. Appl. No. 29/938,392.

* cited by examiner

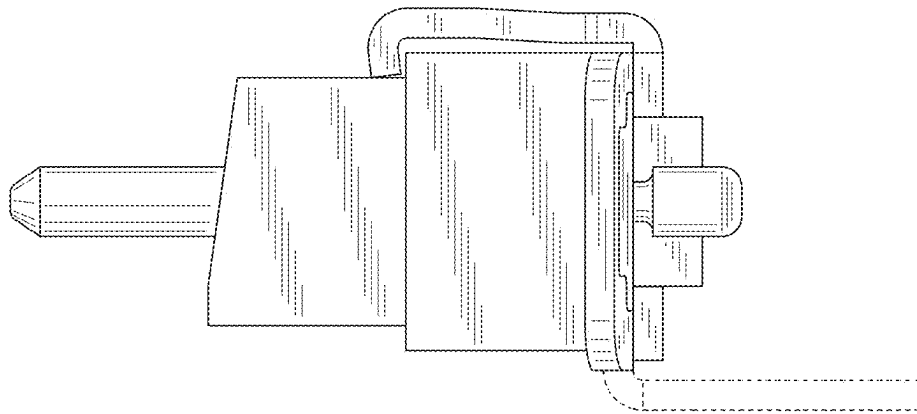


FIG. 1

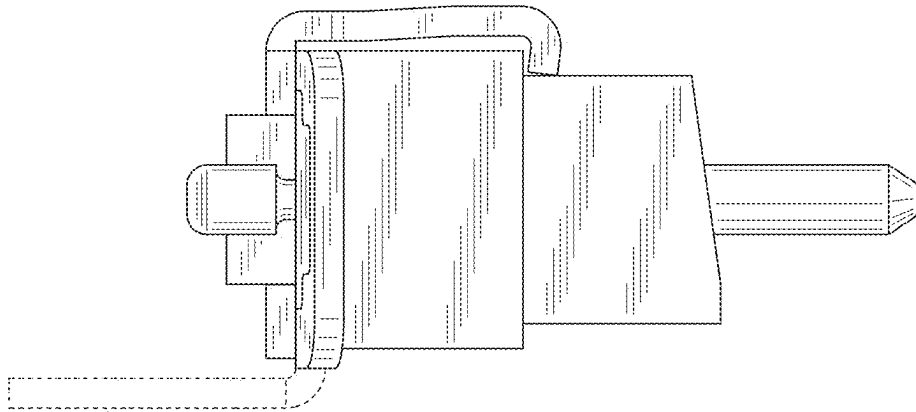


FIG. 2

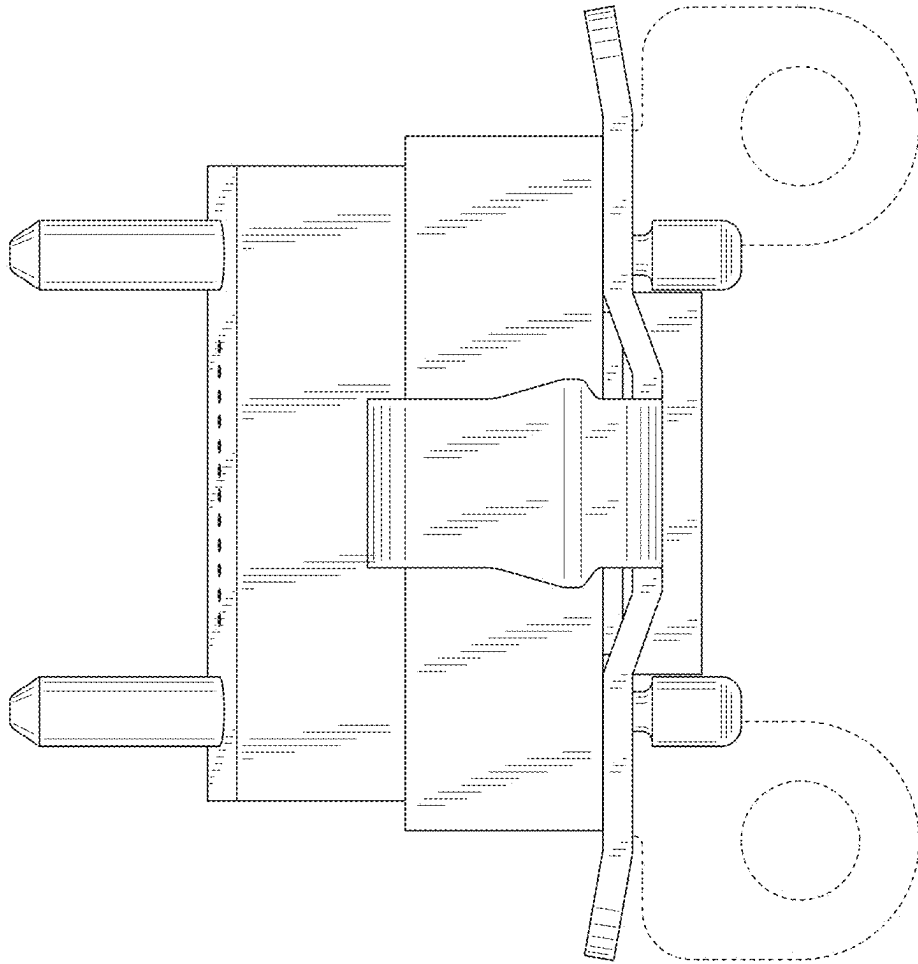


FIG. 3

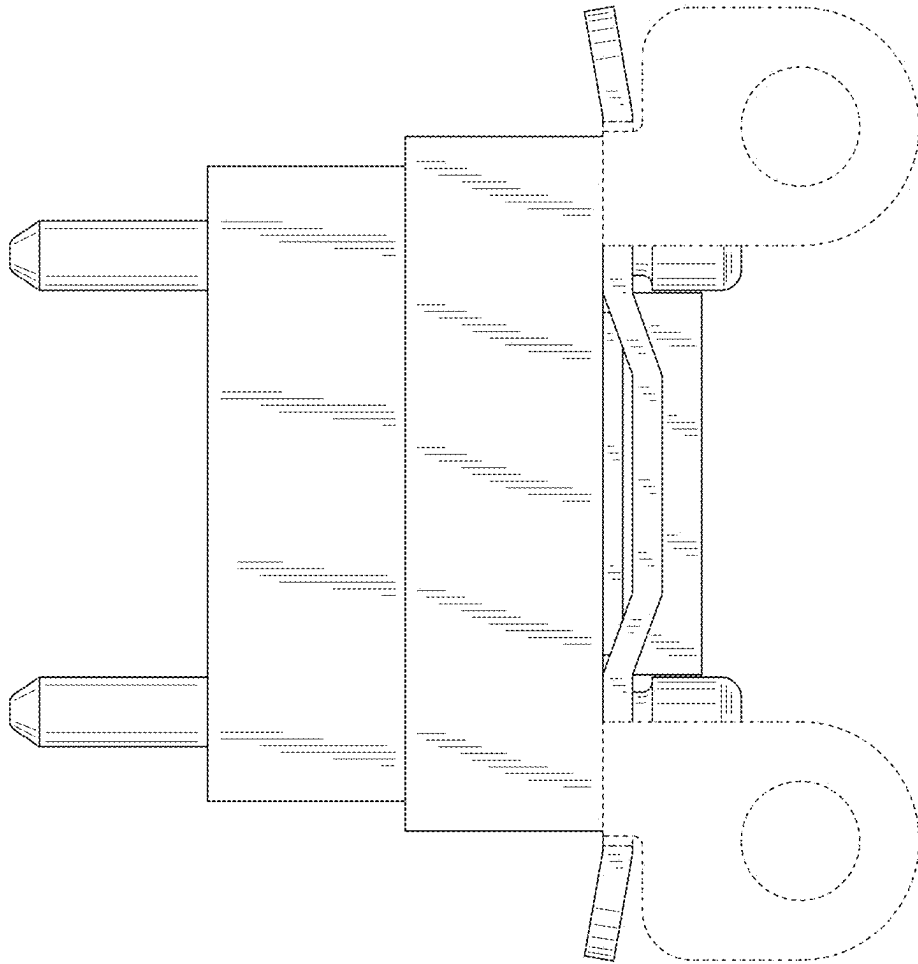


FIG. 4

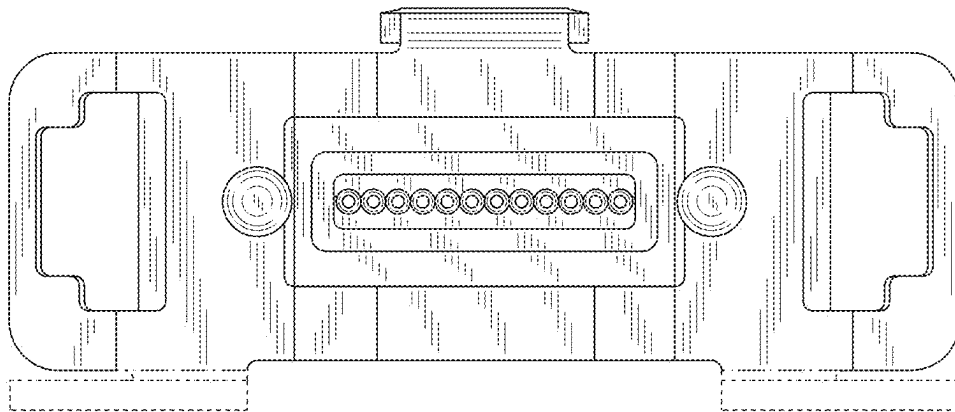


FIG. 5

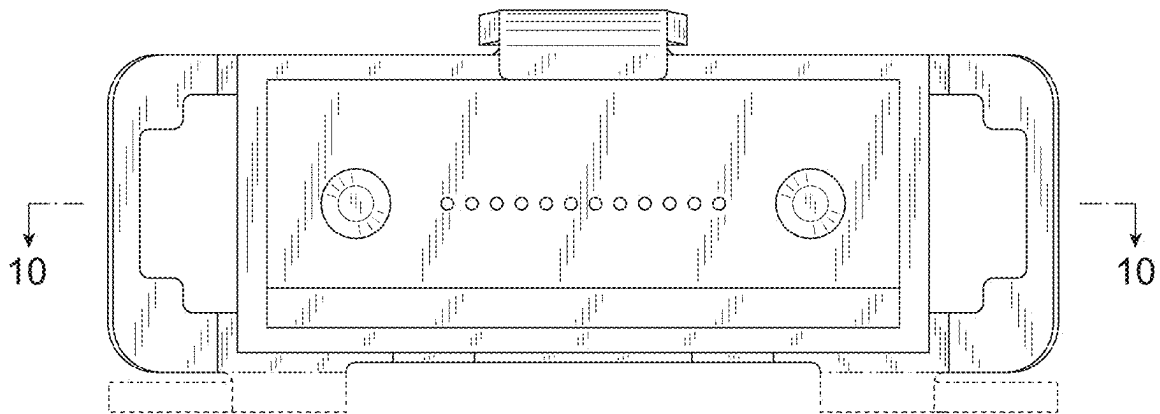


FIG. 6

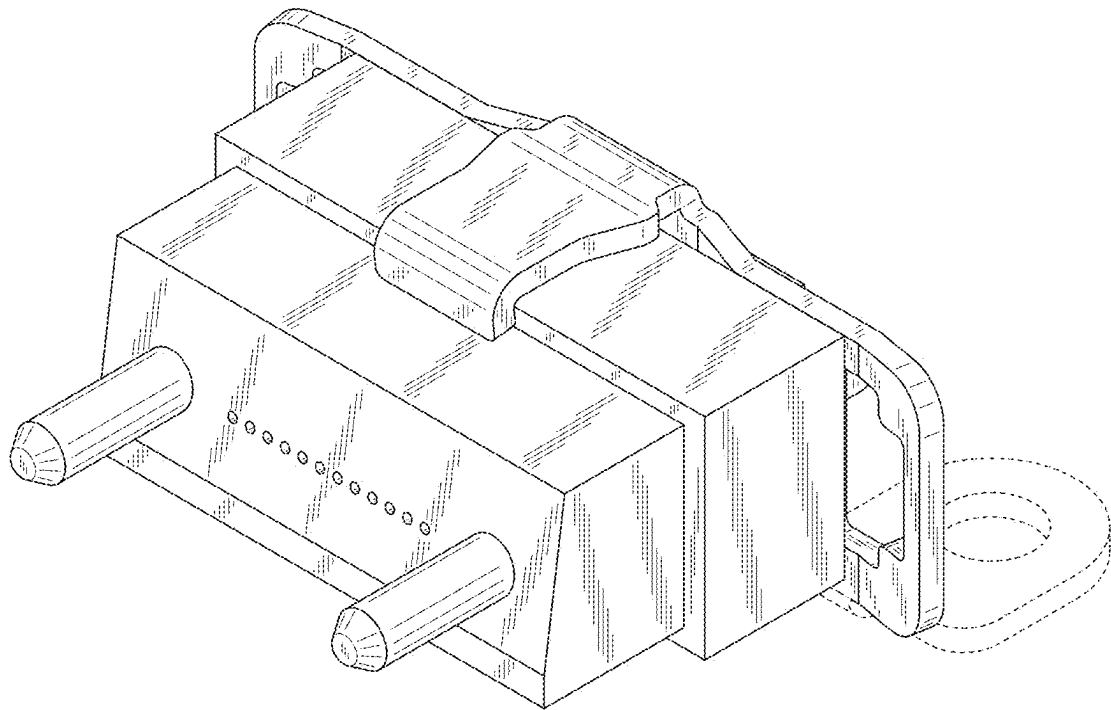


FIG. 7

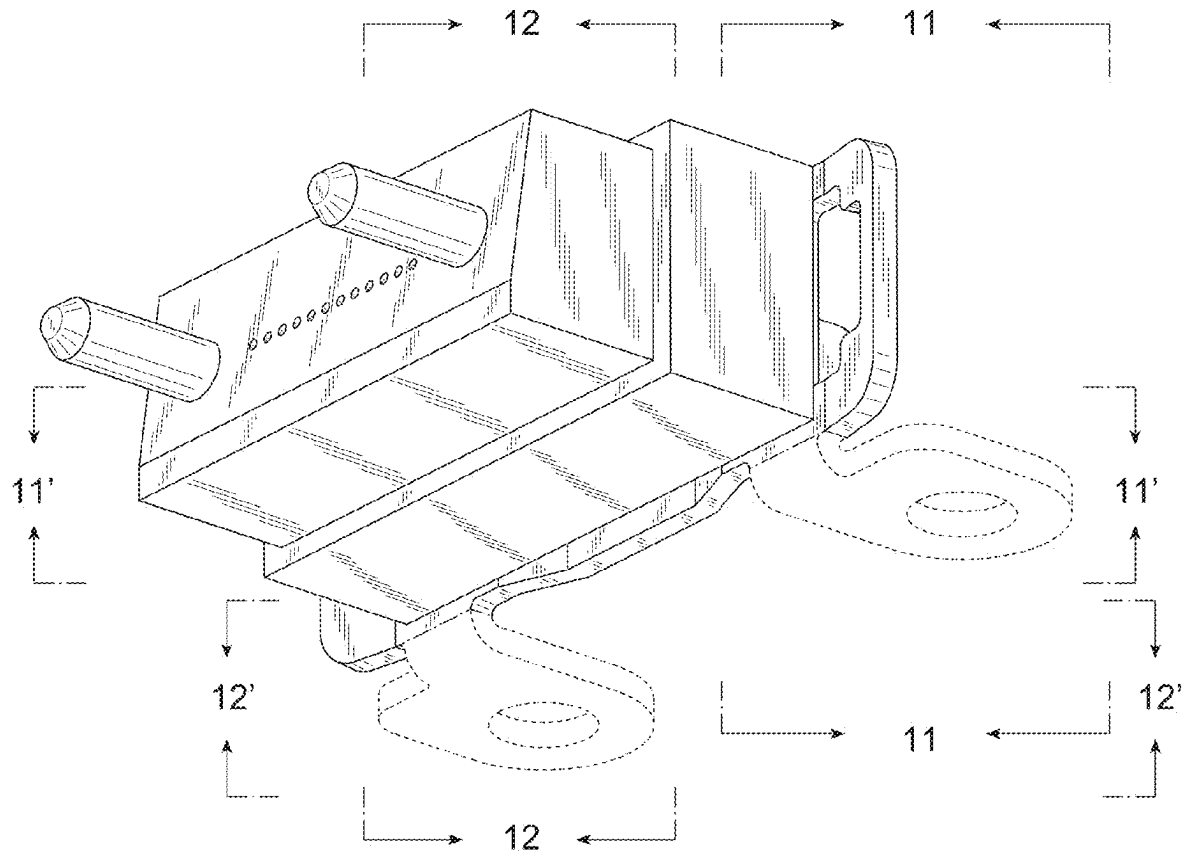


FIG. 8

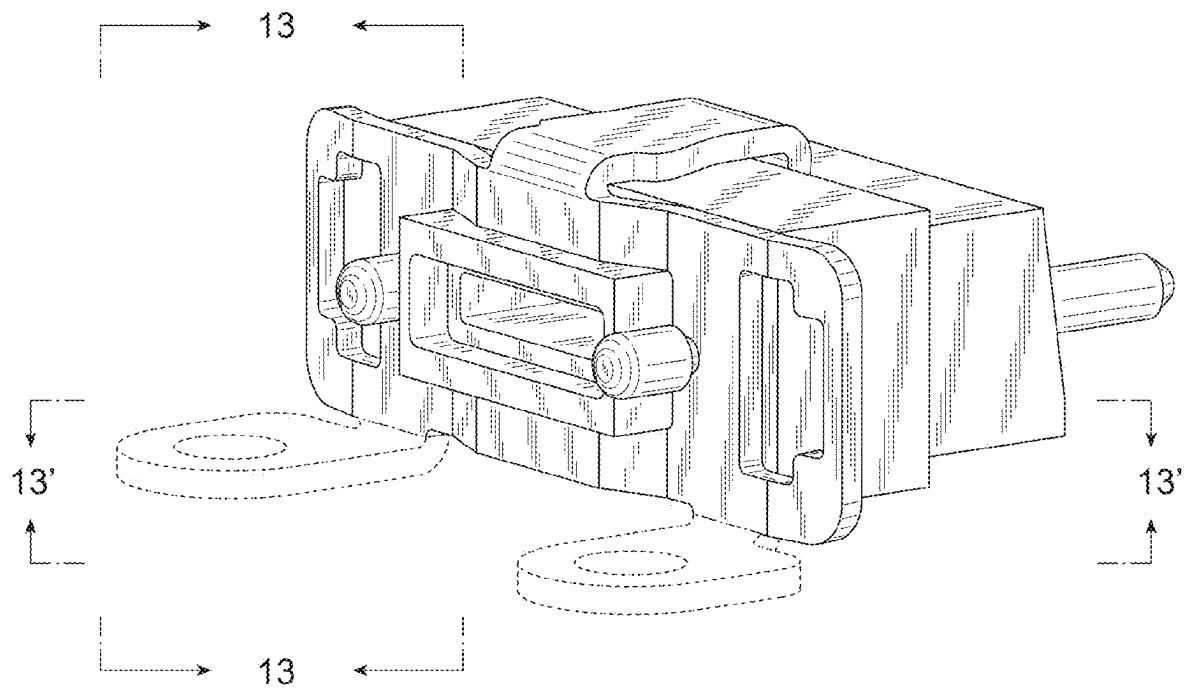


FIG. 9

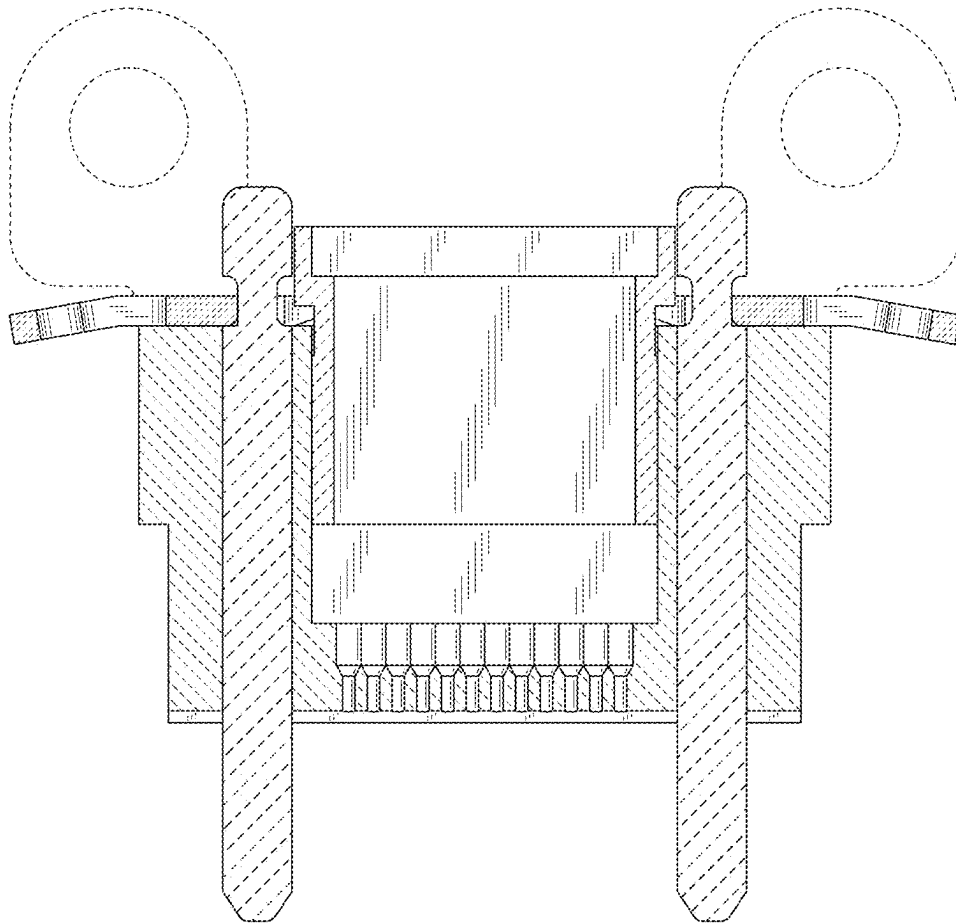


FIG. 10

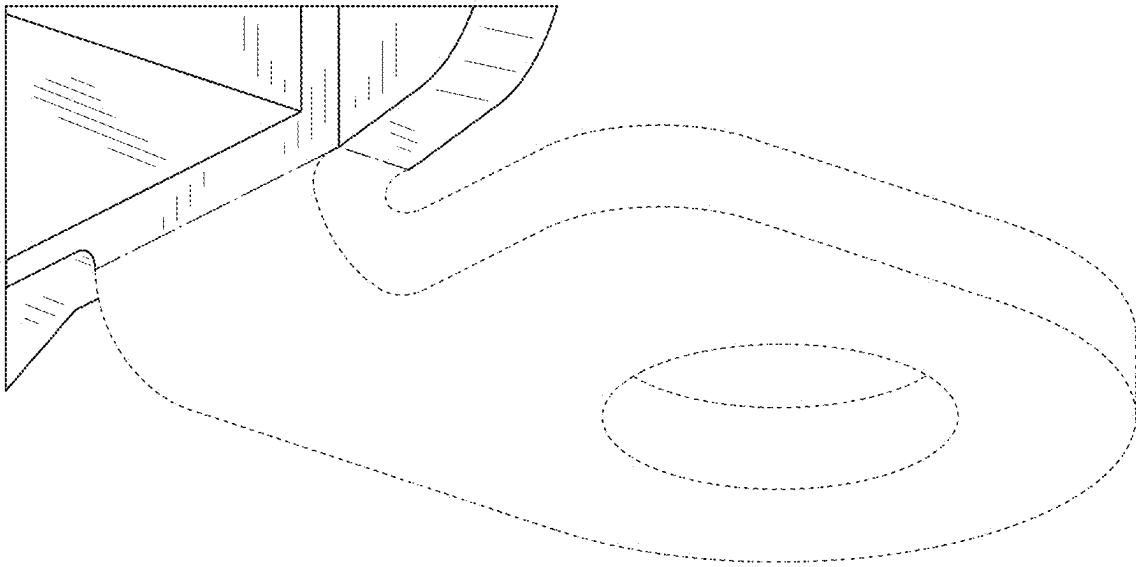


FIG. 11

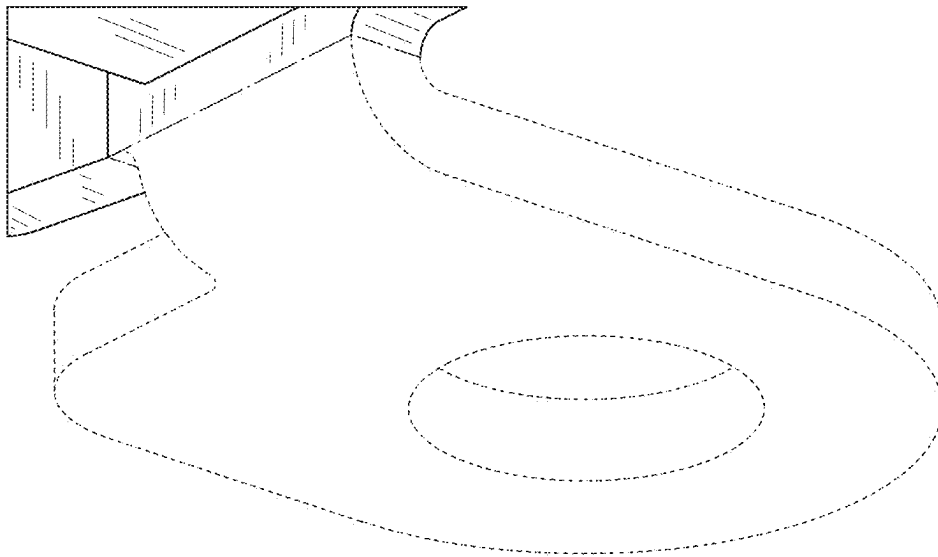


FIG. 12

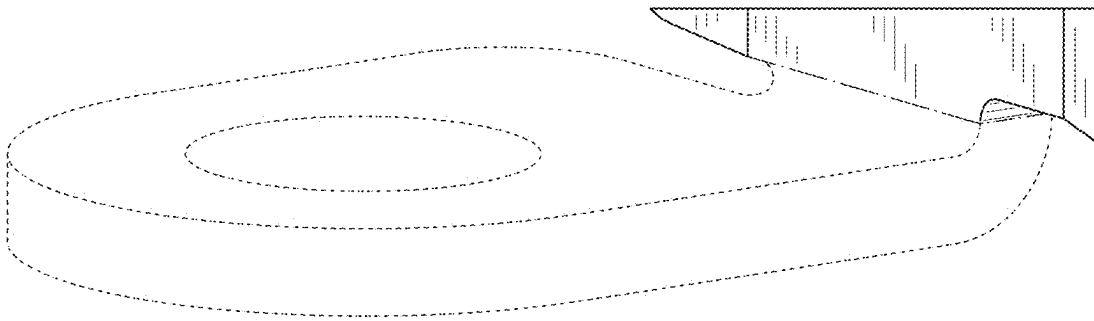


FIG. 13

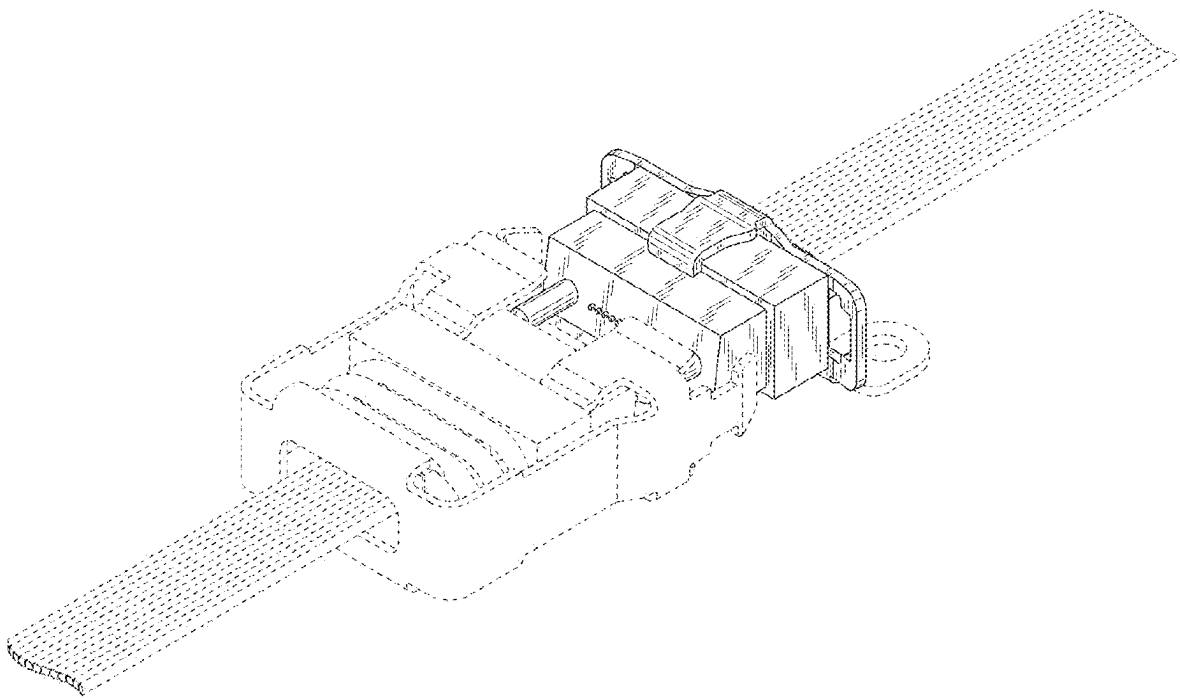


FIG. 14